Problem Set 7

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: CO 4.1–4.2. Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited exercise numbers are from CO (Boyd & Vandenberghe, 2004).

1. CO, Chapter 4 exercises: any three on optimization problems and convex optimization (e.g. Exercises 4.1–4.3).
2. CO, Chapter 4 exercises: one exercise on reformulating a problem into an equivalent convex form.
3. **[code]** Formulate a small convex optimization problem (for example, a constrained least-squares problem) and solve it numerically with a convex solver (e.g. Convex.jl/JuMP in Julia or CVXPY in Python). Report the optimal solution and value, and state which constraints are active at the optimum.
4. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).